

Assignment 1:

a) HOLE

Correctness constraint: print “HOLE” repeatedly, not “HE”, “L”, “O” in any order.

- Process 1: initialize the run, print ‘H’, then wait for process 3 to complete, finish printing ‘E’ and repeat
- Process 2: wait for process 3 to complete
- Process 3: wait for process 1 to pass the lock

Shared states: 3 Semaphores s1, s2, s3

b) HELLO

Correctness constraint: print “HELLO” repeatedly, not “HE”, “L”, “O” in any order.

- Process 1: initialize the run, then wait for process 3 to complete
- Process 2: wait for process 1 to complete, then pass the lock to process 3
- Process 3: wait for process 2 to complete, pass the lock to process 1

Shared states: 3 Semaphores s1, s2, s3

Assignment 2: different threads taking turns to edit a shared variable, using different methods.